



MSc in Computing - Team Project

User Evaluation Report

Group 1 – SmartWardrobe

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1 Introduction

In the current e-commerce environment, consumers' needs are constantly changing, especially in online apparel shopping platforms, where users have extremely stringent experience requirements. With advances in artificial intelligence and image recognition technologies, personalized and interactive shopping features have become a competitive advantage. To meet these demands, our platform integrates innovative features such as virtual try-on, image search, text-based conversational search, and collaborative shopping. However, feature richness must be combined with a quality user experience to truly fulfill user needs. Therefore, we plan to conduct an in-depth user evaluation to ensure the efficiency, ease of use and user satisfaction of the platform's features.

1.1 Why user assessment is needed

In this project, the main objectives of user assessment include:

- **Verify Functional Adaptability:** ensure that the design of AI functions such as virtual try-on and smart search can truly enhance the user's shopping experience and conform to the user's operating habits.
- **Optimize Key Design Elements:** e.g. page loading speed, content quality, visual design, functional layout, etc. to ensure the overall smoothness and visual appeal of UI/UX.

1.2 User Group Definition

The target user group of this project is mainly consumers aged 18-45, who are not only accustomed to online shopping, but also have high expectations for personalized and interactive shopping experience. This group has high acceptance and recognition of emerging technologies such as AI and is willing to try new shopping methods, so features such as virtual try-on and smart search are highly attractive to them. In addition, young users usually want to complete their purchase decisions through fast operation and clear navigation, emphasizing an efficient and intuitive shopping process in a short period of time. Therefore, the definition of user groups in this project also provides a basis for subsequent functional design and user evaluation.

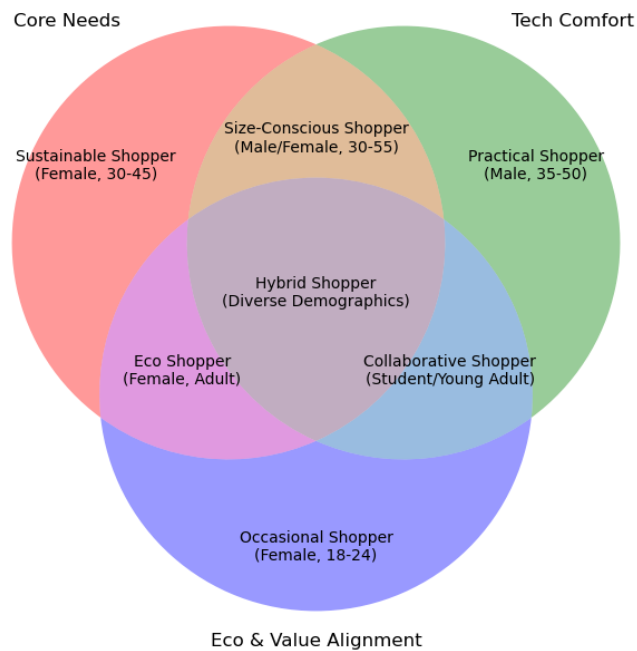


Figure 1: user groups using Venn Diagram

1.3 Functionality Formation Process

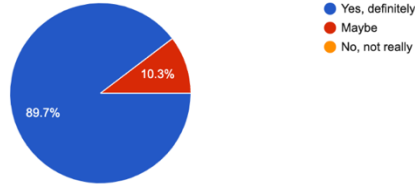
In the process of functionality design, we have utilized the dual feedback mechanism of user surveys and expert reviews to ensure that the design direction is accurate and meets the market demand.

Specifically, we understood users' preferences for online shopping functions through preliminary user surveys, and gradually filtered out differentiated functions such as virtual try-on, image search, collaborative shopping, etc. by combining expert's UI/UX suggestions and horizontal comparisons of existing functions in the industry.

Would you find it useful if a website understood natural language searches and delivered relevant results? (e.g., "Show me casual summer outfits")

29 responses

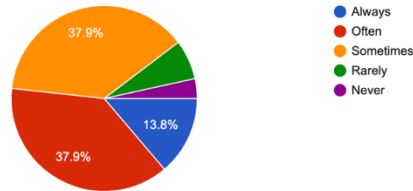
natural language search



How often do you rely on personalized product recommendations while shopping online?

29 responses

Similarity Recommendations



How likely are you to use a feature that lets you upload a picture to find similar products online?

29 responses

image search

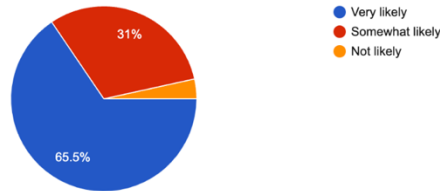


Figure 2: Functional statistics of the initial user survey

Week	Feedback
3	The keyword search used for text search is not as innovative as on other sites, and it is hoped that conversational searches will be added.
4	Conversational search using RAG, add contextual scenarios, LangChain framework; add virtual try-on feature; web theme needs to be adjusted, not in line with fashion apparel shopping site.
5	Add image search function to compare the search results of models horizontally and select the right model.
6	Add collaborative shopping function: friends' conversation, product sharing, etc.
7	The overall UI of the site needs to be adjusted, needs to meet the fashion clothing shopping site design, increase the popularity of elements.

Figure 3: Weekly Progress Report Feedback on Functional Design (week 3-7)

1.4 User evaluation methodology

Based on Nielsen's four-phase evaluation framework In order to scientifically and systematically evaluate the functions and design of the Platform, we have based our evaluation on Nielsen's four-phase methodology for user experience evaluation. In order to scientifically and systematically evaluate the functionality and design of the platform, we based on Nielsen's four-phase user experience evaluation methodology. Below is our specific four-phase plan:

Evaluation Plan Based on Nielsen's Four Phases

Phase	Objective	Activities	Expected Outcome
1. DISCOVER Phase	Identify user requirements and project constraints, analyze the competitive environment, and gather expert feedback	- 1.1 Requirements & Constraints: According to the business objectives and market demand, clarify the functions and interface design requirements that must be realized. - 1.2 User Survey: Through the survey to understand the user's preference and feedback on the virtual try-on, image search and other functions. - 1.3 Competitive Testing: Analyze similar functions, AI applications and user experience features of existing apparel e-commerce platforms , compare the advantages and disadvantages. - 1.4 Expert Review: Obtain suggestions for UI/UX design improvement through expert review.	1.Business Requirements List and User Requirements List; 2. Identify key features and interface elements of the design and analyze the strengths of competing products to optimize the design direction.
2. EXPLORE Phase	Design low-fidelity prototypes based on user personas and scenarios to ensure functionality meets user needs	- 2.1 Persona: Create user personas, including the age, shopping preferences, and technology usage habits of typical users, to guide design decisions. - 2.2 User Stories/Scenarios: Write user stories and scenarios that simulate how a user would accomplish tasks such as shopping and trying on a website, to ensure that the design is close to actual usage scenarios. - 2.3 MindMap: Including key functions such as product search, virtual fitting, collaborative shopping, etc., to ensure clear logic.	1.Identify target users, generate personas; 2.functional mindmap; define user-system interaction flow; 3. generate low-fidelity prototypes, and validate design usability in initial testing.

3. TEST Phase	Test system accessibility, color contrast, and user experience to evaluate design effectiveness	- 3.1 Accessible Checker: Evaluate the accessibility of the design using an accessibility tool. - 3.2 Colour Checker: Use a color contrast tool to check the contrast between the font and the background. - 3.3 Cognitive Walkthrough: Setting up task flow and guiding users to complete it step by step, observing and recording the confusion or obstacles encountered during operation, and identifying usability problems. - 3.4 Qualitative & Quantitative Evaluation: After completing the task, users are asked to rate the usability, visual design, and functional layout of the system (on a 5-point Likert scale) and provide specific feedback.	1. Accessible Design report: Identify and improve accessibility issues in design; 2. Color Checker report: Ensure color contrast is reasonable; 3. Collect user ratings and feedback on interfaces and features.
4. LISTEN Phase	Deeply understand users' emotional feedback, identify potential improvements in user requirements and optimize prototypes	- 4.1 Expert Interviews: Conduct semi-structured interviews with Expert users - 4.2 User Sentiment Analysis: Analyze users' emotional responses in the process of interview and summarize the relationship between emotional feedback and design problems. - 4.3 Updated Medium Fidelity Prototype: Based on the results of the interviews and sentiment analysis, update the medium fidelity prototype.	1. Report on interview results; 2. Obtain emotional responses from users during task execution; 3. Generate mid-fidelity prototypes and refine key design issues.

Figure 4: Evaluation Plan

2 Discover Phase

2.1 Business Requirement

This fashion application's main goal is to develop an immersive, AI-powered online buying experience that is greater than standard e-commerce. Customers can explore, try on, and share fashion products in a distinctive and engaging way using this platform, which blends convenience, personalisation, and social components. The system aims to close the gap between in-store and online shopping by utilising cutting-edge AI-driven features including virtual try-ons, image-based searches, natural language search and collaborative conversation.

2.1.1 Functional Requirements

Account Management:

Log In and Sign Up: Give users the option to sign in to an existing account or create a new one.

Save Favourite Items: This feature improves accessibility and personalisation by allowing users to save items they find interesting for later use.

Product Browsing and Search:

Image-Based Search: Make it simpler for customers to locate particular styles they prefer by allowing them to upload photographs to search for related products.

Text-Based Search: Facilitates a more organic purchasing experience by allowing consumers to type natural language query and receive appropriate search results.

Browse Products: Give customers the option to look at categories, apply filters, and arrange items according to their tastes.

Product Details, Recommendations and Virtual Try-On:

View Product Specifications: Show entire details, including descriptions, size guide, fabric and how it look after me.

Product Recommendation: Let consumers check for similar products from the selected product.

Virtual Try-On: This feature helps customers see products before they buy them by letting them choose a model and see try-on outcomes for specific apparel items.

Shopping Cart and Checkout:

Add to Cart: Give customers the ability to choose how much of an item to put in their shopping cart.

Cart Summary and Checkout: Provide customers with a summary of the products in their cart and walk them through the easy checkout procedure.

Collaborative Chat:

Create a Chat Room: Give users the option to invite friends or relatives to participate in a group chat session inside the app.

Share Product Links: Give customers the option to instantly share product links with friends and family.

Save Chat History: Give consumers the ability to save chat histories so they can review product discussions and choices at a later time.

2.2 User survey

We conducted a user survey to assess whether our identified business requirements align with user needs. A total of 29 users participated, providing 85 responses. After analysing their feedback, we gained the following insights:

2. When searching for products, would you prefer to use natural phrases like "Show me red dresses for summer"?
(29 条回复)

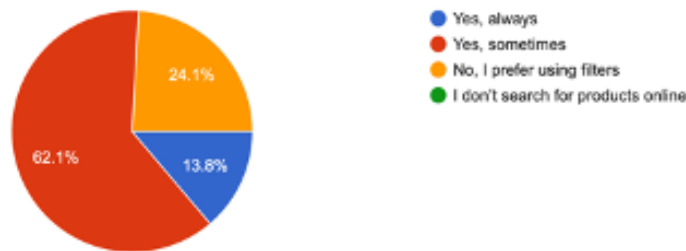


Figure 5: Survey question: would you prefer to use natural phrases

3. Would you find it useful if a website understood natural language searches and delivered relevant results? (e.g., "Show me casual summer outfits")
(29 条回复)

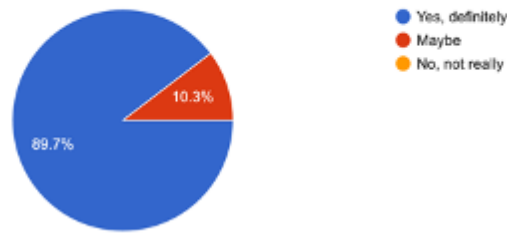


Figure 6: Survey question: it is useful if a website understood natural language searches.

62.1% of users said they only sometimes use natural language when searching, while 24.1% reported using it often. However, nearly all users agreed that a natural language search tool would be useful. This suggests that natural language search holds strong potential to enhance the shopping experience, even if it's not currently top of mind for most people.

15. How likely are you to use a feature that lets you upload a picture to find similar products online?
(29 条回复)

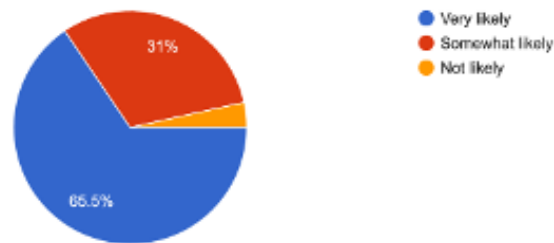


Figure 7: Survey question: How likely user search by uploading an image.

16. Have you ever taken or wished to take a picture to find a product online?
(29 条回复)

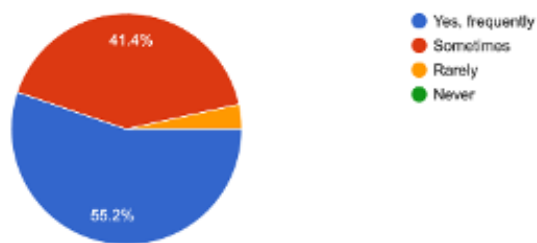


Figure 8: Survey question: How often user search by uploading an image.

About 65.5% of users said they would be very likely to use an image upload tool to find similar products, showing strong interest in this feature. In fact, over 96% have already used it or would like to try it. This suggests that adding image search could really improve the shopping experience, especially for visual items like clothing.

2.3 Competitive Testing

In order to find useful features, user experience improvements, and technical strategies that can improve our own application, competition testing aims to assess and examine comparable systems in the online fashion and retail industry. This analysis aids in obtaining information from well-known platforms that can direct design decisions, feature improvements, and our system's user experience.

2.3.1 Competitor Analysis

Zara (Natural Language Search):

Feature Evaluation: The natural language search function on Zara's website enables visitors to enter questions in conversational terms (such as "red summer dress") and receive relevant results. For users who might not be familiar with particular product names or categories, this tool makes the search process easier.

Insights for Our System: Bringing into practice a natural language processing (NLP) methodology to allow for customisable search terms. More user-friendly experience, customers can search by product qualities like colour, style, and season. Educating the model fashion-specific terms, the search experience can be improved and search results will be more accurate and relevant.

Potential Improvements: To help users get the best results, offer autocomplete options. Adding synonyms and related terms to the search feature can also make it easier for people to find what they're looking for.

John Lewis Fashion Rental (Virtual Try-On):

Feature Evaluation: Before renting or buying, customers can utilise the virtual try-on tool on John Lewis's fashion rental website to evaluate how clothing items would fit them. Customers can sample how products will fit and appear according to this function, which boosts their confidence when they shop online.

Insights for Our System: Using an AI-powered model that can convert apparel photographs to preset model images through a virtual try-on feature. Enabling users to view how clothes suit different body types by choosing different models (based on skin tone, body shape, etc.). Using a realistic rendering model, like VITON-HD, to improve the way texturing and clothing fit are visualised.

Potential Improvements: Providing customisation choices, such the ability to customise the preset models, could improve the user experience even further and make the virtual try-on more engaging.

FindClothesByPicture.com (Image Search):

Feature Evaluation: FindClothesByPicture.com lets users upload a photo and locate related clothing items on the website. Users who want to locate particular products they have seen in person or online but are unable to adequately describe with keywords will find this feature appealing.

Insights for Our System: Creating an image recognition technology that can identify visually similar products by analysing important aspects of uploaded photographs, such as colour, pattern, and clothing type. For precise visual matching of clothing items, convolutional neural networks (CNNs) or comparable AI-based models are used. Concentrating on crucial characteristics like fabric texture, patterns, and colour all of which are crucial for fashion searches in order to ensure high accuracy.

Potential Improvements: Improve the functionality of image searches by adding choices such as "find similar items by colour" or "find similar items by style" to provide more pertinent results. Accuracy could also be increased by offering a feature that allows users to trim or highlight particular areas of an uploaded image.

OuiShopp (Collaborative Chat):

Feature Evaluation: OuiShopp provides a collaborative chat function that lets customers talk about possible purchases with loved ones right on the website. This element enhances the social connection and communality of the shopping experience.

Insights for Our System: Making shopping a shared experience by integrating a chat function that allows users to invite friends or family to participate in a conversation within the app. Allowing users to share product links and private notes on each item using chat features. Including the ability to store chat histories so users can review earlier exchanges and product talks.

Potential Improvements: Provide a smooth transition between chat and product discovery by integrating in-chat recommendations and shopping suggestions depending on the topics being discussed. It would also be more interactive if users could "like" or "recommend" anything in the chat.

2.3.2 Competitive Advantage

Through the integration of features from competitors into a single, integrated platform, our fashion application will offer a distinctive and universal purchasing experience. Modern, fashion-forward customers seeking an interactive and customised online shopping experience will find our focus on user-friendly, AI-enhanced features combined with the social element of collaborative chat to be a one-stop solution.

3 Explore Phase

3.1 Persona building



Name: Harry

Age: 55

Occupation: High school teacher

Goal: Find comfortable and cost-effective products. Quickly find what he need through simple navigation on the website.

Struggles: He doesn't like too many fancy designs on the website and prefers an intuitive and simple interface. If the products recommended by the website do not meet his needs, he may feel that it is a waste of time.



Name: Lisa

Age: 36

Occupation: Full-time mom

Goal : She likes to buy all the items she needs at one time to save time. She likes to use her spare time to browse shopping websites and add the items she likes to her wish list so that she can compare them when shopping.

Struggles: The website page loading speed is slow, and the shopping experience is not smooth. When buying clothes online, she can't see how the clothes will look on her. It is very troublesome to return the clothes after she gets them home.



Name: Lily Watson

Age: 25

Occupation: Marketing Specialist

Goal: Be able to quickly find fashionable clothing that suits her taste. Hope shopping website can help her find clothes for different scenarios, such as daily life, office and vocation.

Struggles: Online shopping is not like shopping with friends offline, where you can communicate with each other in real time, making the shopping experience more enjoyable. Referring to the outfit sharing on social media, she hopes that the website can help her find clothes similar to the ones in the picture.

3.2 User Stories/Scenarios

User Account Management:

Scenario: I want to register as a new user so that I can save my favourite things, and quickly manage my profile.

Motive: Having an account enables me to retrieve my saved products more rapidly, and customise my profile for recommendations in the future.

Browsing Products:

Scenario: To find the things I'm looking for quickly, as a user I want to be able to browse products by categories and filter by price.

Motive: Since I might not have a specific purchase in mind, browsing allows me to see everything the website has to offer. Sorting and filters make it easier and faster to find things that fit my preferences by reducing the number of options.

Text-Based Product Search:

Scenario: In order to get precisely what I'm looking for as a user, I wish to search for products using keywords or natural language queries (for example, "summer blue dress").

Motive: Even if I don't know the actual product name, it's simple to find products given to a flexible, natural search tool. This lessens the time and effort required to find particular items in a huge catalogue.

Image-Based Product Search:

Scenario: As a user, I intend to upload a picture of a product I like so that the website can display related products.

Motive: There are times when I want to purchase something but find it difficult to put into words. Shopping becomes easier when I upload an image because I may see similar things without knowing specific search terms.

Viewing Product Details:

Scenario: Before adding an item to my cart as a user, I would like to see comprehensive product descriptions, size guide, fabric, related items, and availability.

Motive: Being aware of every detail about a product, such as its colours, sizes, and suggested complementary items, helps me make well-informed purchases and lowers the possibility of returns.

Virtual Try-On:

Scenario: As a user, I would like to visually put on clothes to see how they might fit me using a predefined model because I don't want my image to be uploaded in a random website.

Motive: A virtual try-on tool helps me see how a product might fit, which boosts my confidence in my purchasing decision and improves the overall shopping experience because I can't physically try on items when I shop online.

Collaborative Chat:

Scenario: As a user, I would like to set up a chat room where friends and family can discuss product URLs and details.

Motive: Getting advice from others is helpful when selecting stylish products, and shopping is frequently a social activity. I can make better decisions with the help of collaborative discussion and the advice of people I can trust.

Add to Cart and Checkout:

Scenario: As a user, I like to review a summary of my choices, add products to my cart, and finish my transaction via a safe checkout procedure.

Motive: It's easy and convenient to add products to a cart and check out, which lets me check my purchases before making a purchase. I can be sure that my order and information are handled securely when the checkout process is quick and easy.

Saving Favourites:

Scenario: As a user, I would like to add items that catch my attention to a favourites list so that I can easily retrieve and compare them later.

Motive: Since I might not be ready to purchase something right now, keeping track of things enables me to review my options and make a decision later.

3.3 Mind Map

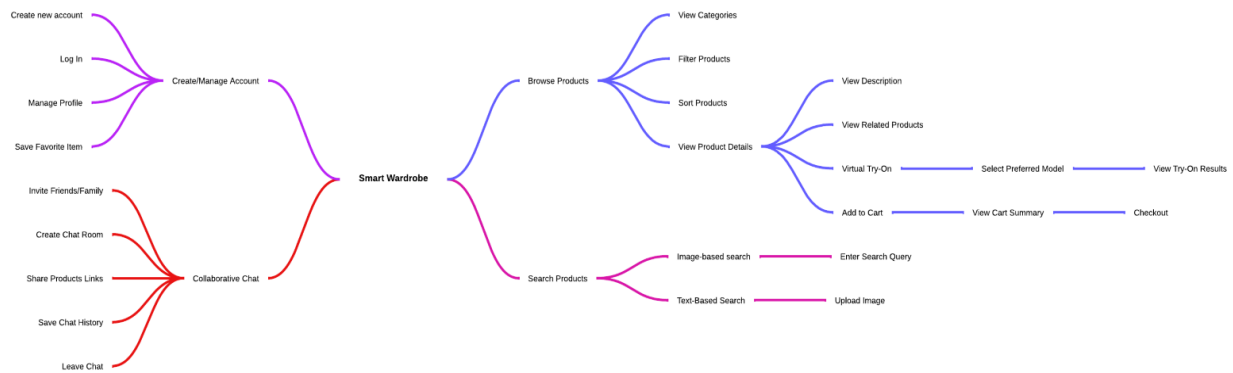


Figure 9: Mind Map

3.4 Low fidelity prototype

Our team designed the first version of a low-fidelity prototype, which was a hand-drawn sketch.

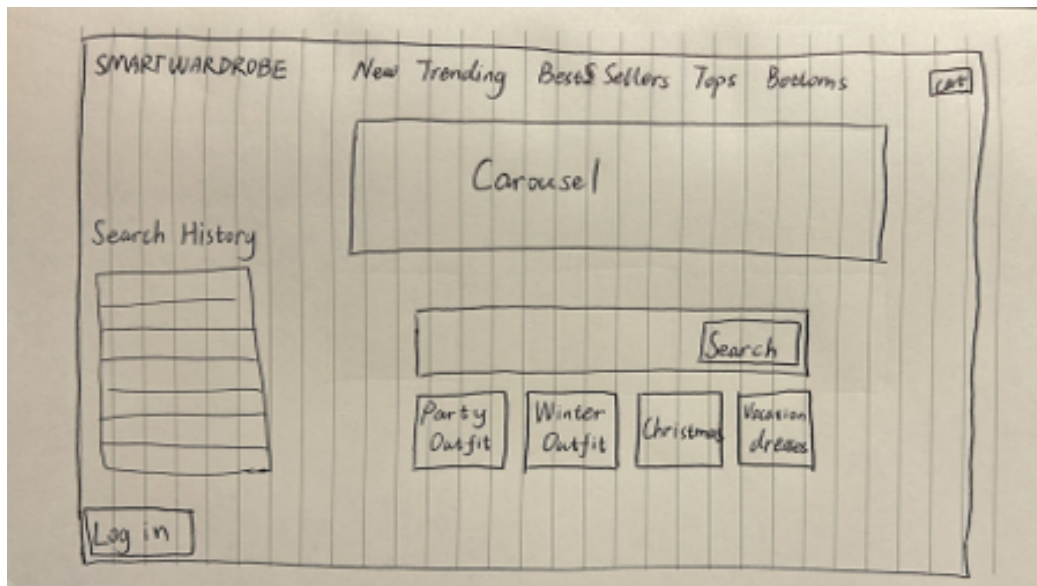


Figure 10: First Version of Low-fidelity Prototype - Home page

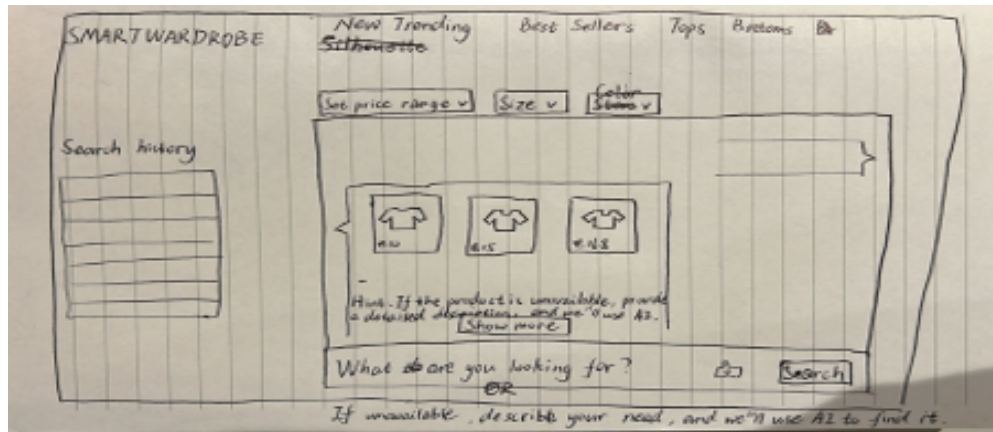


Figure 11: First Version of Low-fidelity Prototype - Products Page

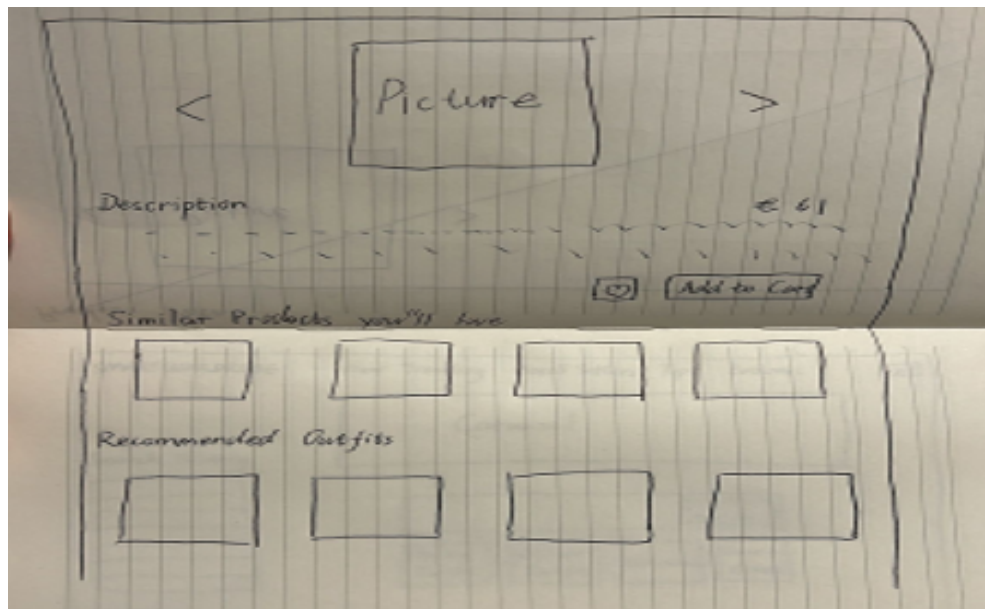


Figure 12: First Version of Low-fidelity Prototype - Product Details page

We conducted user interviews with the first version of low-fidelity prototype in hand. The users provided feedback on the design, including suggestions for the 'log in' button and search bar. So we moved 'log in' button from left bottom to right top, and moved the search bar from the bottom of page to top. Then we made the second version of low-fidelity prototype.

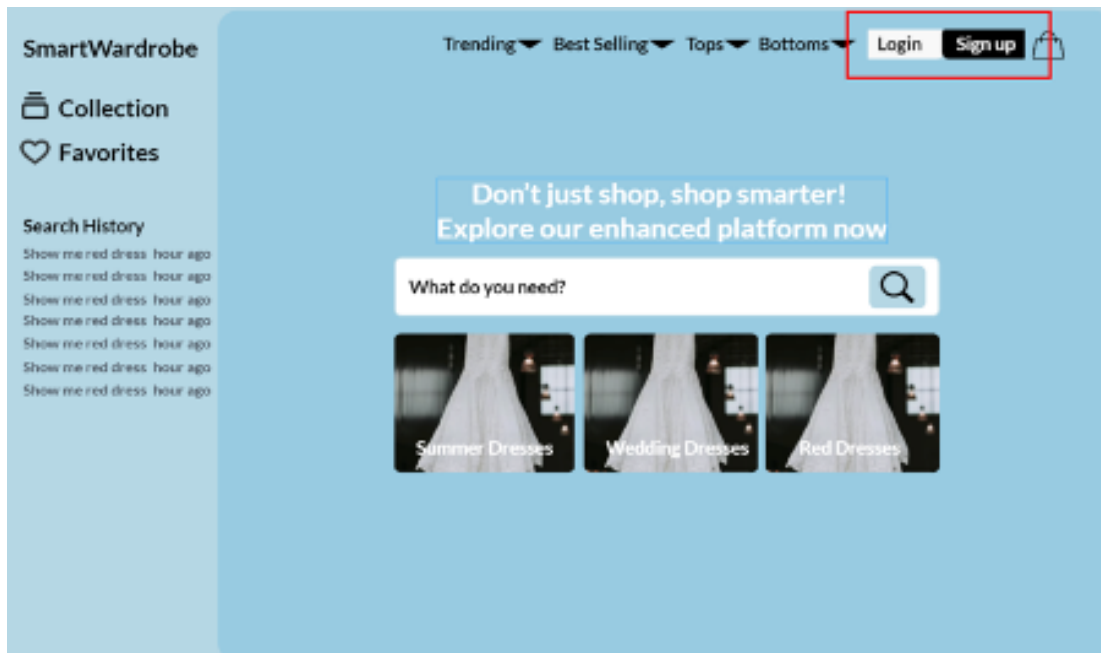


Figure 13: Second Version of Low-fidelity - Home page

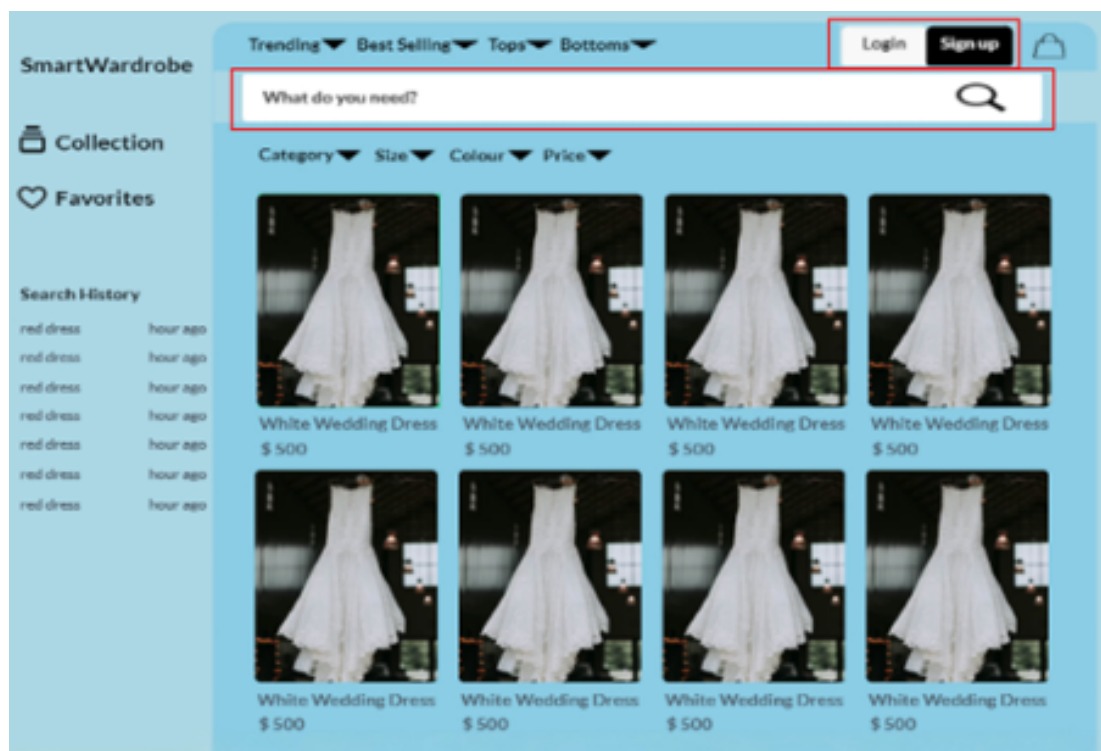


Figure 14: Second Version of Low-fidelity - Products page

We presented the second version of our low-fidelity prototype to experts in software development, who are also our project tutors. The experts noted that our current design didn't fully align with user expectations for a fashion website, especially in terms of color coordination. They suggested

that we start displaying products on the homepage or provide more options to enhance interactivity. With this feedback, we redesigned the user interface and created the following new mock-ups.

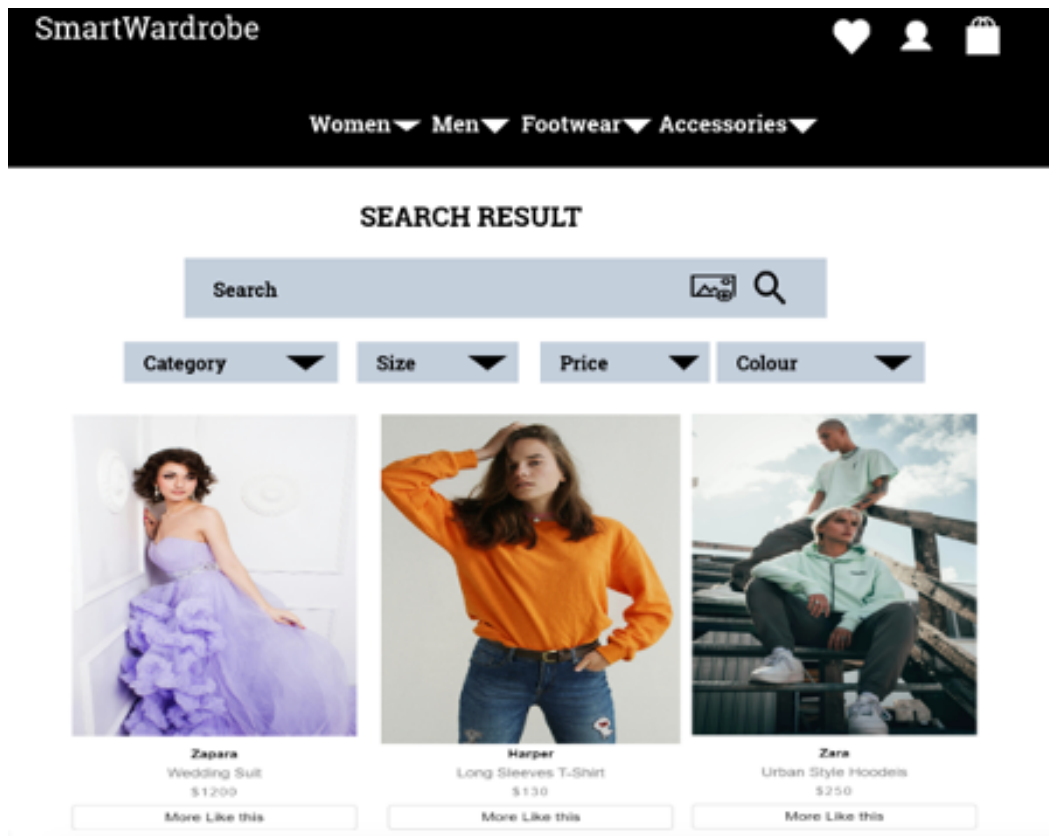


Figure 15: Redesign Mock-up - Products Page

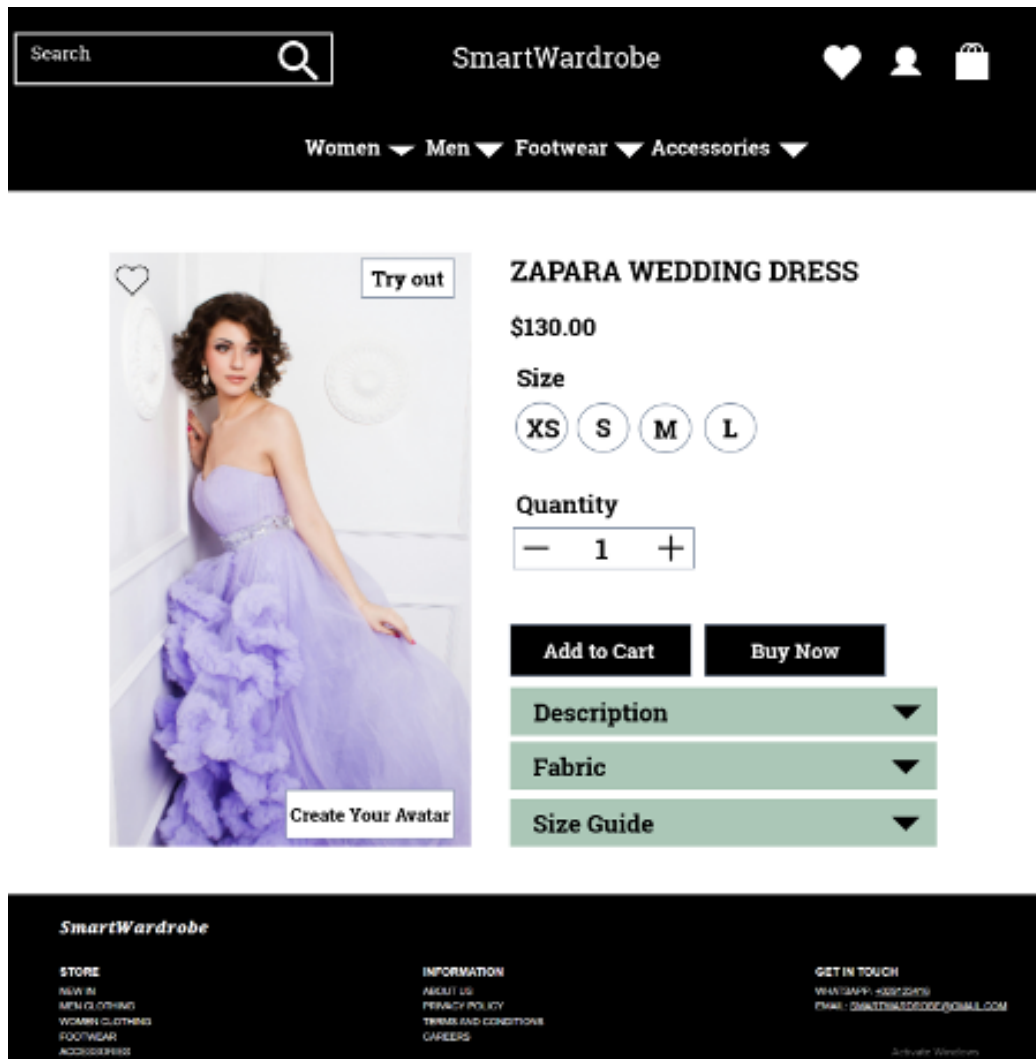


Figure 16: Redesign Mock-up - Products Details Page

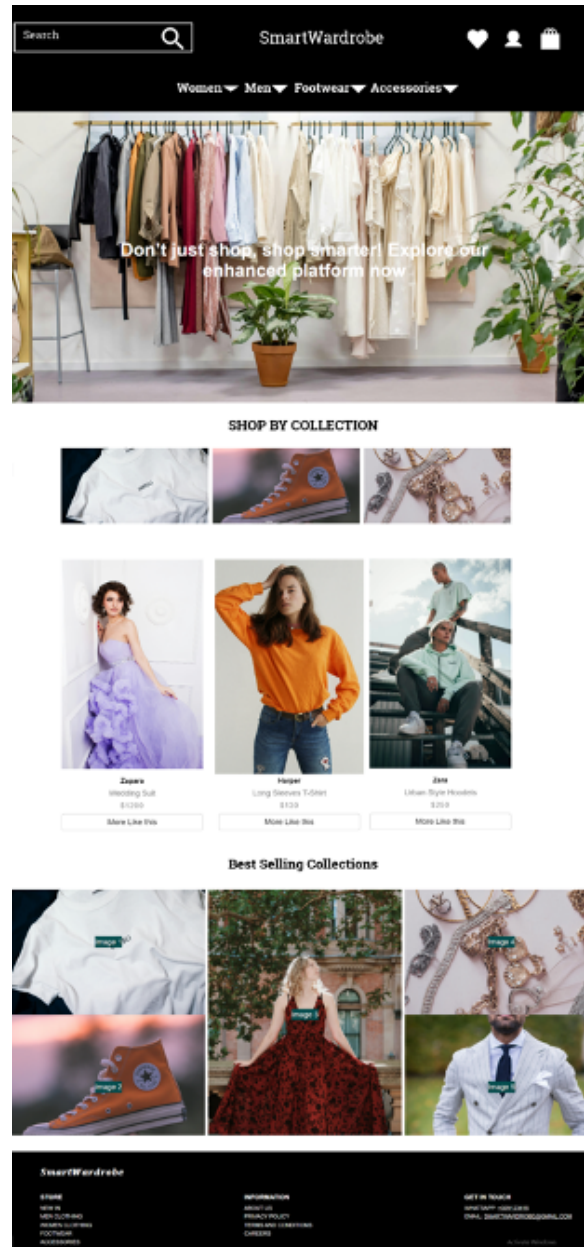


Figure 17: Redesign Mock-up - Home Page

In terms of functionality, experts suggested adding a virtual try-on feature to our website. After receiving this suggestion, we reached out to users who had previously indicated in our survey that they shop online weekly, and we interviewed them to gather their thoughts on the virtual try-on feature. The users found the idea fresh and exciting, noting that being able to see how clothes might look on them before purchasing would greatly enhance their shopping experience.

We then asked if they would be willing to upload their own photos to the website to create a virtual try-on image. Some users expressed interest in seeing how the clothes would look on themselves,

while others were concerned about privacy issues. After identifying these user needs, we created a mock up of the virtual try-on interface.

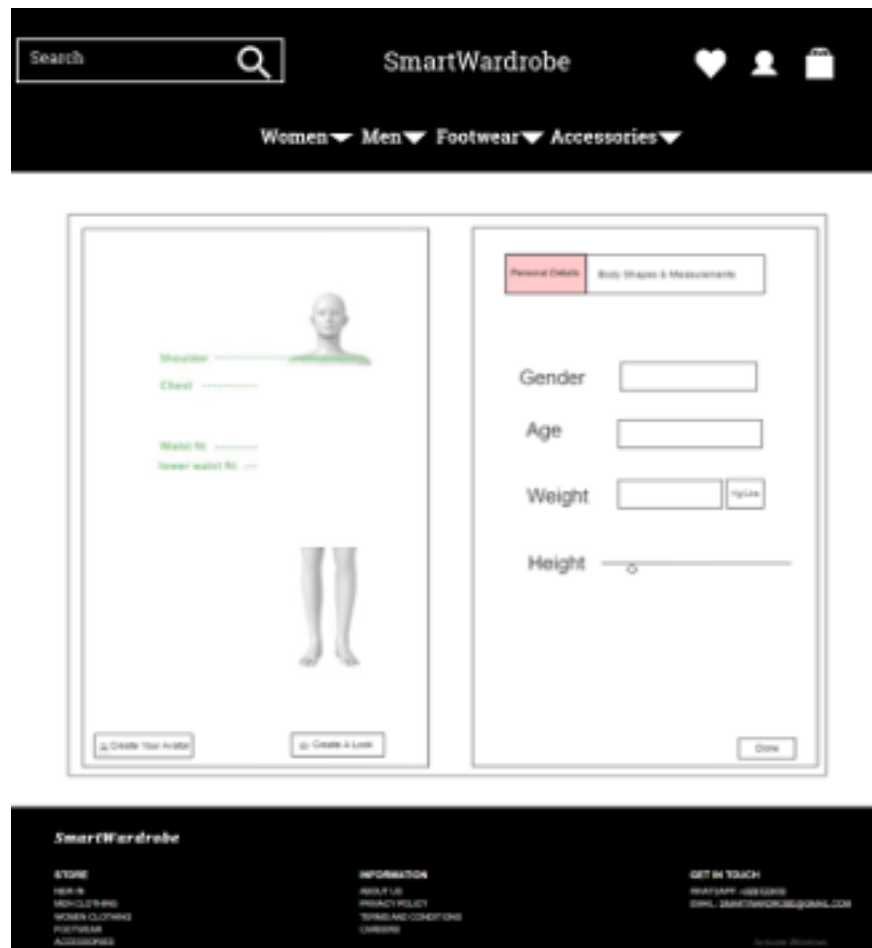


Figure 18: Virtual try on page mock-up

This design protects user privacy by allowing users to customize a mannequin model's body dimensions rather than uploading their own photos. The website then generates a virtual try-on preview using the mannequin model alongside the clothing image. When we presented this design to experts, they appreciated the privacy-protection approach but found the appearance of the mannequin model unsatisfactory and suggested improvements. With this expert feedback, we redesigned the virtual try-on interface and created the following mock-up.

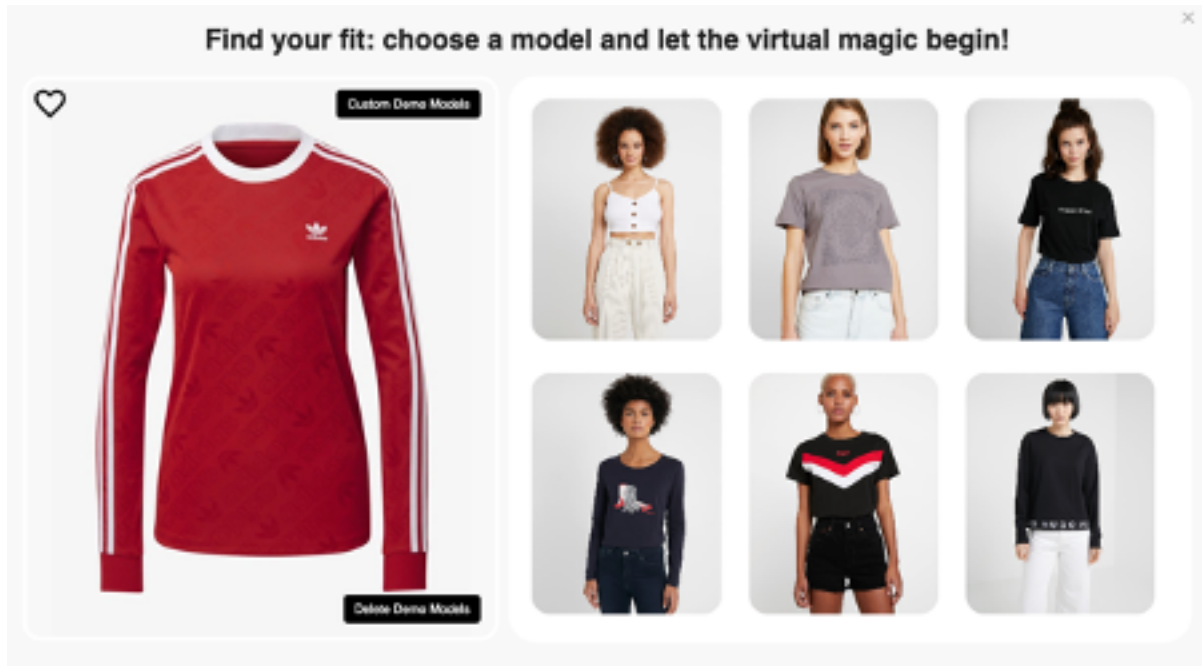


Figure 19: Virtual try on page

4 Test Phase

4.1 Accessible testing

The Web Content Accessibility Guidelines (WCAG) are the primary reference in our user evaluation process to ensure that websites are accessible to all users, including those with different visual, auditory, or cognitive abilities. The WCAG standards focus on four main principles: Perceivable, Operable, Understandable, and Robust. Perceivable), Operable, Understandable, and Robust. The following accessibility and color detection tools were used for this test

Tool 1: WAVE Web Accessibility Evaluation Tools

WAVE is an accessibility testing tool that focuses on detecting common accessibility issues in web pages. It clearly presents accessibility issues in web pages through visual annotations, including testing for image substitution text, page structure, links, forms, etc. - Scope of testing: WAVE tests the compliance of page structure, alternative text, keyboard navigation, form labels and link text.

Procedure:

1. Scan the page through WAVE for a report containing accessibility issues.
2. Specific issues are examined one by one according to their location, such as whether page titles are missing, whether images have alternative text, and whether form labels are complete.
3. Based on the accessibility issues detected, and fix them according to the tool's recommendations.

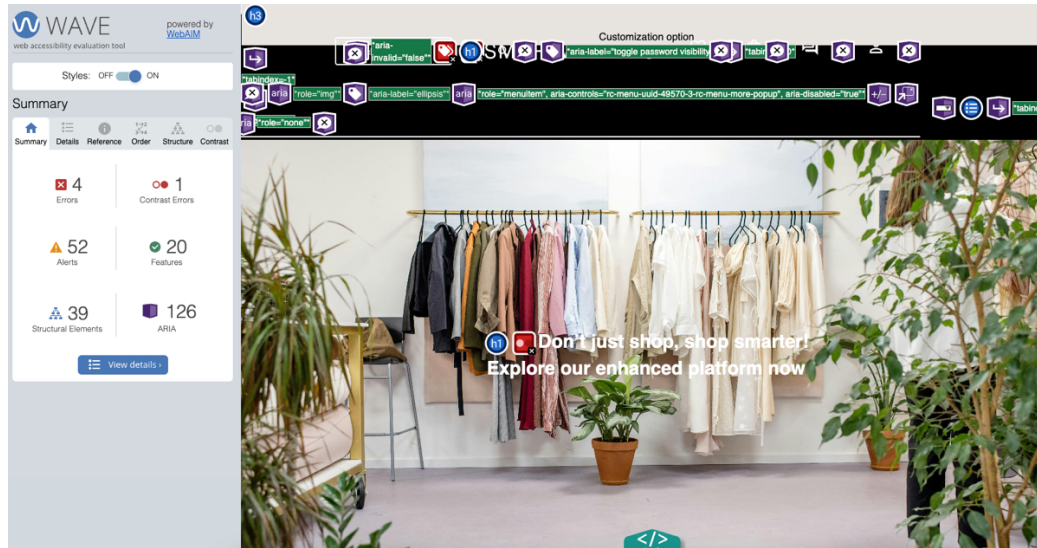


Figure 20: Example of reported results prior to accessibility testing

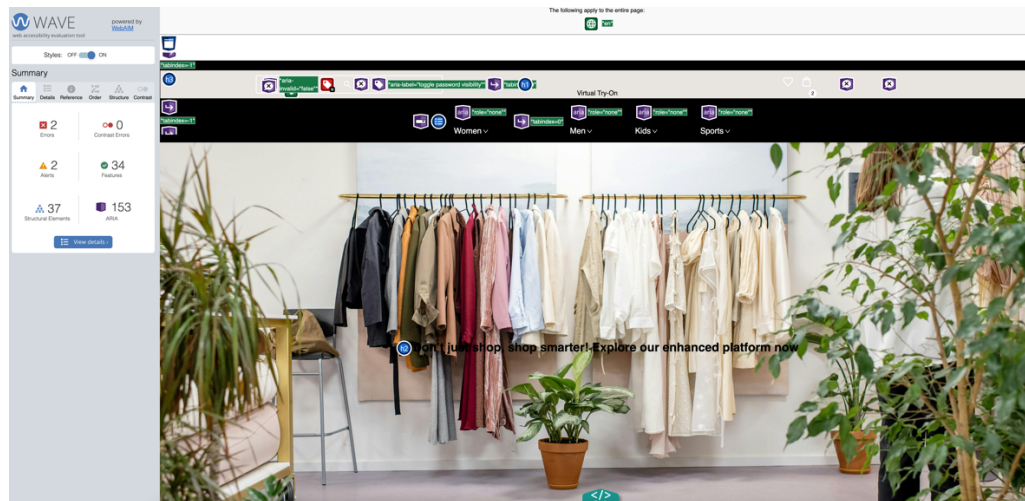


Figure 21: Example of optimized accessibility test report results

Tool 2: IBM Equal Access Accessibility Checker

IBM Equal Access Accessibility Checker is a browser-based plug-in tool that provides in-depth inspection of web page accessibility, including complex interactions and JavaScript-driven dynamic content.

The tool detects HTML structure issues, color contrast, missing alternative text, and generates specific accessibility reports for subsequent fixes and improvements.

Scope: IBM Equal Access Accessibility Checker mainly covers HTML structure, keyboard navigation, use of ARIA tags, contrast issues, form accessibility and other aspects.

Steps to perform:

1. Enable IBM Equal Access Accessibility Checker in the browser and scan the page.
2. Review each of the scan results for issues such as button and link legibility, form field fit, ARIA tag usage.
3. Based on the accessibility issues detected and fixed according to the tool's recommendations.

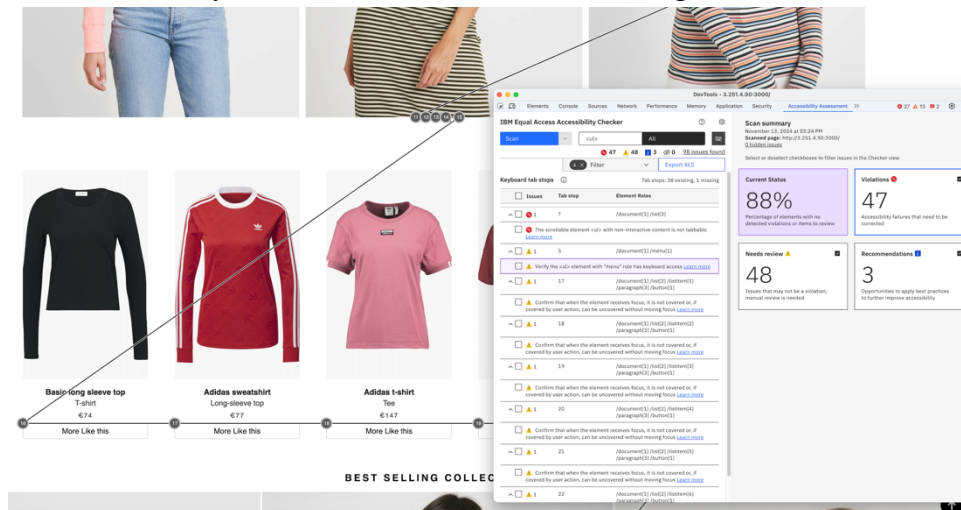


Figure 22: Example of reported results prior to accessibility testing

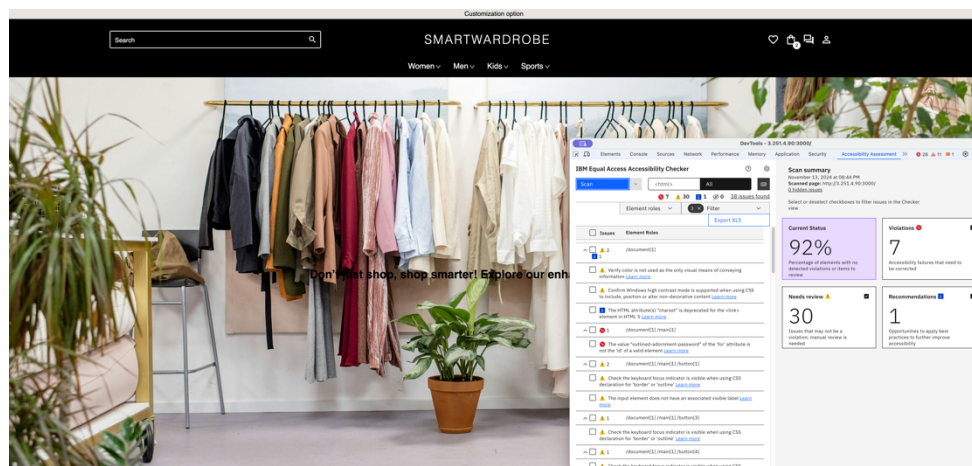


Figure 23: Example of optimized accessibility test report results

4.2 Colour checker

Tool: WCAG Color contrast checker

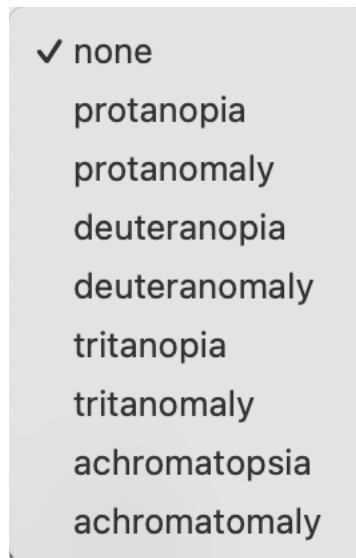


Figure 24: Types of Color Blindness: protanopia; protanomaly; deutanopia; deuteranomaly; tritanopia; tritanomaly; achromatopsia; achromatomaly.

To ensure that all users, especially those who are colorblind, are able to correctly identify visual elements on a page, we use the WCAG Color Contrast Checker tool.

This tool is tested for the following types of color blindness: protanopia, protanomaly, deutanopia, deuteranomaly, tritanopia, tritanomaly, achromatopsia, and achromatomaly. with the tool, we were able to Make real-time contrast adjustments during the design process to ensure that minimum contrast requirements are met (e.g., no less than 4.5:1 text to background).

Scope of testing: Primarily used to check color contrast to ensure readability of different color combinations.

Procedure:

1. The WCAG Color Contrast Checker is used to check each color combination on a page. 2. Different types of color blindness are simulated to ensure that all users can identify key information and action buttons on the page. 3. Changes are made to non-compliant color combinations to ensure that the color contrast complies with the WCAG standard.

Tests show that the color contrast of the current website is compliant and no further adjustments are required.

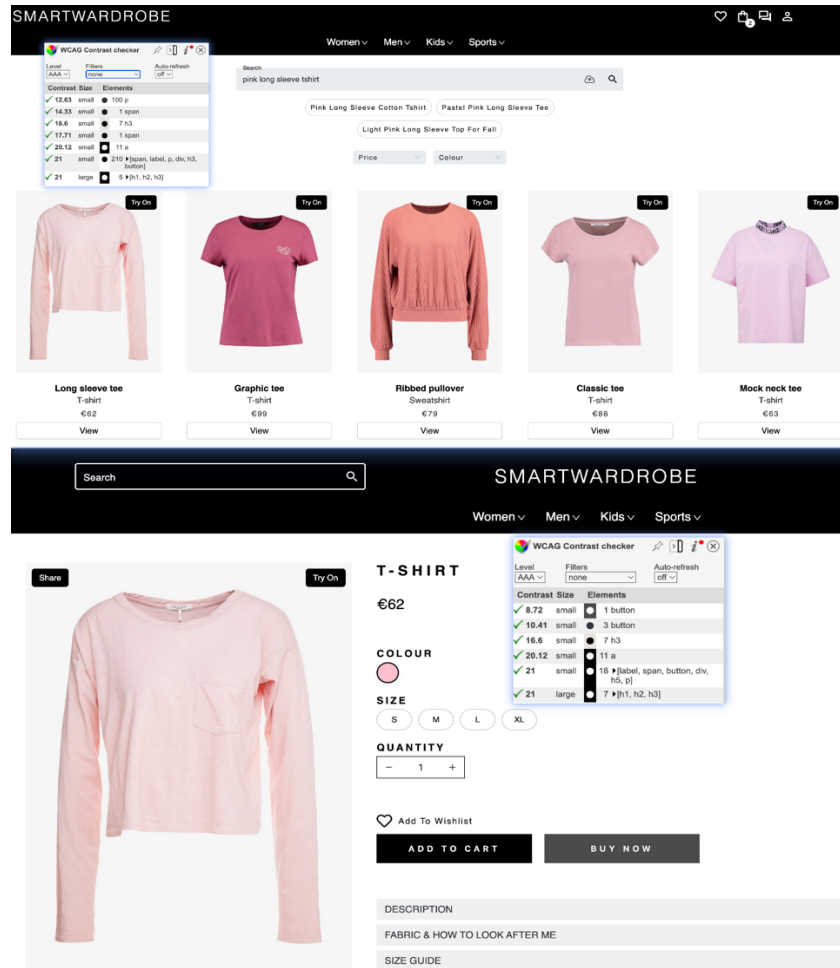


Figure 25: Examples of normal vision conditions

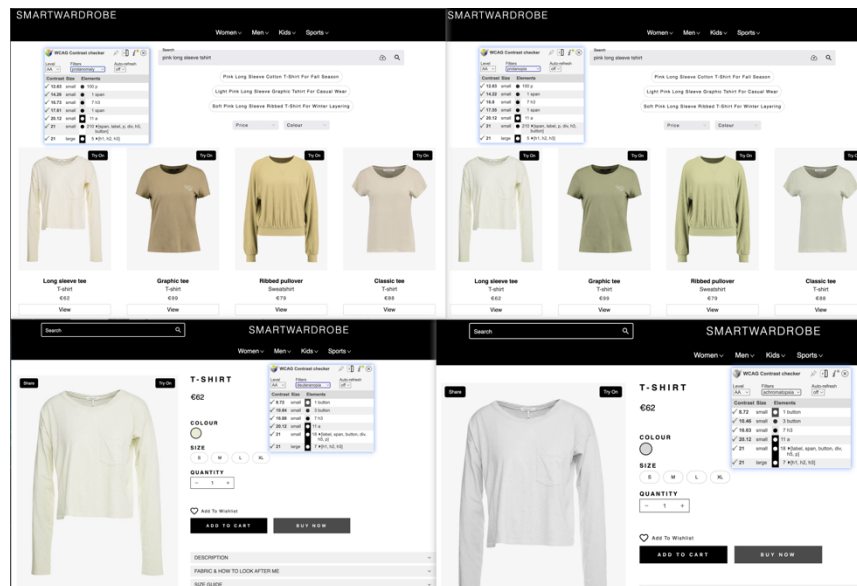


Figure 26: Examples of other vision conditions

4.3 Medium fidelity prototype experimental method

4.3.1 Experiment design

In evaluating the medium fidelity prototype, we used a combination of Cognitive Walkthrough, Qualitative Questionnaire, and Quantitative Questionnaire methods.

1. *Cognitive Walkthrough*

The Cognitive Walkthrough is an analytical method used to assess the usability of a user interface (Polson, Lewis, Rieman, & Wharton, 1992). It is particularly useful for evaluating the experience of first-time users of Smart Wardrobe, because it allows us to observe how effectively users can complete tasks within the interface. By evaluating the task completion process, we can identify any potential design flaws. Once the core functions of the product are implemented, the Cognitive Walkthrough helps us verify whether the design aligns with actual user needs. This evaluation can reveal gaps between the design and user expectations. We can make adjustments in an early stage. Observing users' interactions with Smart Wardrobe provides the team with valuable insights into users' thought processes, allowing us to refine interface details and enhance the overall user experience.

We designed 22 user tasks to complete within the Smart Wardrobe. The team will release tasks one by one to users. Users are asked to complete these tasks as they naturally would in their everyday browsing habits. The tasks cover all core functionalities of the Smart Wardrobe website. **Below are the user tasks we designed:**

- 1 Can you perform sign up on this website?
- 2 Can you perform log out on this website?
- 3 Can you log in on this website, by using account created in task 1?
- 4 Can you use search bar to perform search? Like you input something in search bar and search.
- 5 Can you filter the currently searched products within a certain price range(20-50)? Can you skip filtering by price range to see all search results? Can you filter products by a specific color?
- 6 Do you know the function of the three lines of text below the search bar? Can you use this feature to perform a search?
- 7 Can you upload an image and try to find products similar to the given image?
- 8 Can you add a friend named "Jiaxin" in the collaborative chat interface to share the shopping experience?
- 9 Can you send the message "hi" to your friend "Jiaxin" in the collaborative chat?
- 10 Can you exit the collaborative chat interface and randomly share a product link with your friend Jiaxin?
- 11 Can you open the collaborative chat interface, start a group chat, and invite "Jiaxin" to join the group chat?

- 12 Can you view details of a specific product? (Including: description, fabric, size guide)
- 13 Can you add this product to wishlist?
- 14 Can you view your wishlist?
- 15 Can you remove item from your wishlist?
- 16 Can you choose a product and try it on a preset virtual model? Can you put this product on other models?
- 17 Can you find similar products and virtually try them on models?
- 18 Can you add one product in your cart?
- 19 Can you view your cart?
- 20 Can you modify the quantity of products in cart?
- 21 Can you remove items in your cart?
- 22 Can you find this company contact information?

After we get user tasks result, we need to determine whether the user successfully completed each task. **The conditions for success are:**

- a. The user completes the task goal successfully without relying on additional help.
- b. The user makes no significant errors in the task, or makes only a few minor errors that are corrected, and successfully completes the task.
- c. The user completes the task as expected, without having to take detours or return multiple times.
- d. The user completes the task within a reasonable amount of time.

If any one of the conditions is "no", the task is considered a failure.

2. Qualitative Questionnaire

We designed 12 questions to collect Qualitative data. After users complete all the above tasks, we will collect their feedback in the form of interviews.

A. On the overall website experience

- 1 How was your overall experience when browsing and using the website? Did you encounter any confusing or unclear operation steps?
- 2 How would you rate the interface layout and visual design of the website? Are there any design elements that you are particularly satisfied or dissatisfied with?

B. Virtual Try-on Function

- 3 How did you find the Virtual Try-on? Did it help you to better select the right style?
- 4 Did you find the Virtual Try-on smooth and easy to understand? Were there any areas of the fitting process that were not clear or needed improvement?

C. Image Search and Text Search

5 Do you find the image search function accurate? After uploading an image, did you get search results similar to it?

6 Do the results of the search box meet your expectations? Does the Associative Search Button generated below the search box give you search convenience?

D. Collaborative shopping and friend interaction

7 Do you feel that the collaborative chat room feature would have a positive impact on your shopping process? (selection and purchase of items, buying more stuff, making purchase decisions faster) . How smooth is the process of interacting with friends and sharing products?

8 Would you like to see more features in the collaborative chat room? (Besides sending messages, sharing products) .

E. Wish List and Shopping Cart

9 Are you satisfied with the experience of using the Wish List? Was it easy to add and delete items?

10 Do the shopping cart features meet your expectations? Is the operation of adding and deleting products smooth?

F. Suggestions for Improvement

11 What do you think needs to be improved in the existing functions and design of this website to enhance your shopping experience?

12 What other features do you think could be added to help you select and purchase items more easily?

3. Quantitate Questionnaire

Based on Page Load, Content Quality, Visual Design & Accessibility, Functionality, User Journey, Responsiveness, Navigability, and User-Friendliness, we designed 14 questions in the questionnaire. We hope that through the targeted answers from users, we can identify the advantages and room for improvement of smart wardrobe and provide direction for further optimization. The content of the questionnaire is placed in the appendix.

4.3.2 Experiment implementation

In selecting participants for the experiment, we referred to the previously identified user groups and personas. Before starting the experiment, we inform participants that the session will be audio- and screen-recorded; these recordings will be used solely for academic analysis, and we obtain the users' consent for this. The experiment takes place on the participants' personal laptops. We ask each user to open our website on their own laptop and complete all assigned tasks. After completing all tasks, we conduct an interview with each participant to collect qualitative data, followed by a survey to gather quantitative data.

4.3.3 Experiment data analysis

1. Cognitive Walkthrough

S: success

F: fail

Task score: success times/attempt times(The number of times this task has been executed)

	T1	T2	T3	T4	T5	T6	T7	T8	T9	T10	T11
User 1	S	S	S	S	S	F	S	F	S	S	S
User 2	S	S	S	S	F	S	F	F	S	F	S
User 3	F	S	S	S	S	S	F	F	S	S	S
User 4	S	S	S	S	S	S	S	F	S	S	S
User 5	S	S	S	S	S	F	F	F	S	F	S
Task Score	80%	100%	100%	100%	80%	60%	40%	0%	100%	60%	100%

	T12	T13	T14	T15	T16	T17	T18	T19	T20	T21	T22
User 1	S	S	S	S	S	F	S	S	S	S	S
User 2	S	S	S	S	S	S	S	S	S	S	S
User 3	S	F	S	S	S	S	S	S	S	F	S
User 4	S	F	S	S	S	S	S	S	S	S	S
User 5	S	S	S	S	S	S	S	S	S	S	S
Task Score	100%	60%	100%	100%	100%	80%	100%	100%	100%	80%	100%

Figure 27: Task score: success times/attempt times

The data we've collected so far reveals the following prominent issues:

- **T8:** The *Add a Friend* feature is hidden within a dropdown menu, and none of the test participants located the button immediately.
- **T7:** The image search feature is not in the homepage search bar, causing some users to miss it. Additionally, when users hover over an uploaded image to preview it, the preview covers the delete button, making it difficult to remove the image.
- **T6:** Users were eventually able to deduce the function of the *generated query search* button after reading, but it took them some time. We could improve the design by clearly indicating that these are clickable buttons that will initiate a search automatically.
- **T10:** Users encountered difficulty locating the *Share* button.
- **T13:** Users spent a considerable amount of time searching for the *Add to Wishlist* button.
- **T1:** The *Create Account* button is not prominent, causing users to spend extra time finding it.
- **T5:** One user attempted to select “white” in the color filter, but there was no option for white.
- **T17:** The *Similar Products* section is located at the bottom of the virtual try-on page, and some users did not notice it.
- **T21:** On one user's laptop, the delete button was not displayed in the cart.

2. Qualitative Questionnaire

Questionnaire Qualitative question	Questions	Feedbacks		
A. On the overall website experience	Q1, Q2	The user found the overall experience positive, describing the website as user-friendly and easy to navigate, even for a new user. The user noted some unclear operation steps, specifically mentioning the login page. They suggested making the “Like” and “Register” buttons more distinct for better clarity. A design improvement was suggested for the search bar, recommending a brighter background color (e.g., white) to make it more noticeable.	The user found the website interactive and appreciated the option to demo clothes on different body types, mentioning this feature demonstrates inclusivity. Navigation was described as efficient and user-friendly, with necessary product information easily accessible for adding items to the cart.	The user found the browsing experience smooth but noted that all products are not available (e.g., only tops are present). They found some buttons not intuitively placed and suggested that buttons be aligned with the product description and potentially placed at the bottom instead of overlapping with images. The user rated the interface layout an 8 out of 10, noting
B. Virtual Try-on Function	Q3, Q4	The user was satisfied with the virtual try-on feature, finding it helpful for selecting the right style. They appreciated this unique feature, as it is rarely seen on other websites. Suggestions included adding male models and expanding the try-on options to include full outfits, such as trousers. The user also mentioned wanting to see the model's specific measurements to better understand the fit. Include model size information to help users better gauge the fit.	The user praised the try-on feature, particularly the ability to preview items on models of various body types. They noted this is a unique feature, rarely seen on other websites, and appreciated the inclusivity it promotes.	The user had not previously experienced a similar feature but found it potentially helpful for envisioning how clothing items look on real people. They suggested incorporating models with diverse body shapes, including overweight models, as the current models all share a similar, slim body type. The virtual try-on process was described as smooth and easy to understand.
C. Image Search and Text Search	Q5, Q6	The user found the image search function accurate, noting that it displayed items with similar colors and styles. They appreciated the generated search suggestions based on input, as it provided useful inspiration for further exploration. The user felt that three suggestion options were sufficient, stating that more would be overwhelming.	The user found the image search and text search features useful but suggested making the image upload option more accessible. They recommended placing an “upload image” button directly within the search interface to streamline the process and reduce user interactions.	The user noted that the image search was not entirely accurate, possibly affected by the presence of many yellow colors in the background of the uploaded image. They suggested enhancing the image search function and expressed a preference for categorical options in search suggestions — breaking down search queries by gender, color, and style for more tailored options. The generated associative search buttons did not meet their
D. Collaborative shopping and friend interaction	Q7, Q8	The user liked the collaborative chat room feature, noting it as a unique and useful addition that allows friends to discuss items together. They suggested implementing friend recommendations based on similar style preferences or creating style-based communities (e.g., a group for black t-shirt enthusiasts).	The user viewed the collaborative chat feature positively, noting that it is convenient for discussing products with friends directly on the website. They highlighted that this eliminates the need to switch to another app for sharing product links, which enhances the shopping experience.	While the chat room feature could potentially be useful, the user noted that its practicality is situation-dependent. They suggested integrating voice chat to enhance communication efficiency. They found the overall interaction with friends smooth but emphasized that voice communication could speed up the
E. Wish List and Shopping Cart	Q9, Q10	The user was satisfied with the overall experience of the wish list and shopping cart, describing them as clear and easy to use. They suggested adding an edit option within the cart to change the color or size of items without removing them first. The user recommended displaying the item count for both the wish list and shopping cart in a consistent location on the page.		The user felt the “Add to Wishlist” button was too small to notice immediately but found adding or deleting items easy. They requested a feature to sort items in the wishlist by categories like T-shirt, pants, casual, or formal, especially valuable if the wishlist grows large. The shopping cart mostly met expectations, though they
F. Suggestions for Improvement	Q11, Q12		The user mentioned the importance of keyboard accessibility, such as incorporating “Enter” and “Escape” keys for smoother navigation, as they frequently use these keys.	The user recommended expanding clothing options and improving the ability to adjust item size within the cart.

Figure 29: Feedback from Qualitative Questionnaire(Detailed information is listed in the appendix)

Based on the qualitative feedback provided, the analysis of areas for improvement in the UX/UI:

A. On the overall website experience

Clarify operation steps, particularly on the login page, and make function buttons like "Like" and "Register" more distinct. Improve the visual distinction of elements (e.g., using more than just black, white, and gray) to clearly differentiate buttons and navigation options.

B. Virtual Try-on Function

Introduce models with diverse body shapes, including options for male and plus-size models. Provide model-specific measurements to help users better gauge how clothes might fit.

C. Image Search and Text Search

Make the image upload feature more accessible within the search interface. Enhance the accuracy of the image search function, considering background colors that might affect results. Offer more categorical options in search suggestions for tailored user experiences.

D. Collaborative shopping and friend interaction

Integrate voice and possibly video chat options to improve communication efficiency. Consider developing style-based communities for more personalized friend interactions.

E. Wish List and Shopping Cart

Enlarge the "Add to Wishlist" button for better visibility. Implement options to sort wish list items by categories (such as type of clothing). Allow edits to item colors and sizes in the cart without requiring removal and re-addition.

F. Suggestions for Improvement

General Recommendations:
Enhance keyboard accessibility, allowing for smoother navigation with keys like "Enter" and "Escape."
Expand clothing options and facilitate adjustments of item sizes within the cart.

3. Quantitate Questionnaire

Metric Measured	Question Index	Measuring Criteria (Success)	Measuring Criteria (Failure)
Page Load	Q1	≥ 4	< 4
Content Quality	Q2, Q3	≥ 4	< 4
Visual Design & Accessibility	Q4, Q5, Q6, Q7	≥ 4	< 4
Functionality	Q8, Q9	≥ 4	< 4
Responsiveness	Q10, Q11	≥ 4	< 4
Navigability	Q12	≥ 4	< 4
User-Friendliness	Q13, Q14	≥ 4	< 4

Figure 30: Measuring Criteria for Quantitate Questionnaire

	Page Load	Content Quality		Visual Design & Accessibility				Functionality		Responsiveness		Navigability	User-Friendliness	
User	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	Q11	Q12	Q13	Q14
u1	5	4	5	5	3	3	4	4	5	5	5	5	5	5
u2	4	4	5	4	3	4	5	3	5	4	5	4	4	3
u3	4	3	2	4	2	3	4	3	4	3	3	3	4	4
u4	2	5	5	4	4	4	5	4	5	3	2	5	5	4
u5	4	5	4	5	4	5	5	4	5	3	4	4	4	5
u6														
Average Score	3.8	4.2	4.2	4.4	3.2	3.8	4.6	3.6	4.8	3.6	3.8	4.2	4.4	4.2
Success Score	80.00%	80.00%	80.00%	100.00%	40.00%	60.00%	100.00%	60.00%	100.00%	40.00%	60.00%	80.00%	100.00%	80.00%

Figure 31: Feedback from Quantitate Questionnaire

The following information is derived from the results of the above quantitative analysis:

A. Page Load: While the success score is relatively high, the average score of 3.8 indicates that users are generally satisfied but there is room for improvement in page load speed to ensure users do not experience delays.

B. Content Quality: Users find the product descriptions detailed and accurate, and the complementary visual and textual information on product pages appear clear. Maintaining consistently high-quality content will keep supporting user satisfaction.

C. Visual Design & Accessibility: While most users find the design aesthetically pleasing and icons intuitive, text readability (Q5) receives a significantly lower score. Focusing on improving font size and color contrast can enhance readability and visual accessibility, addressing this key pain point.

D. Functionality: The overall shopping process is seamless, but some users struggle to find necessary functions easily. This suggests potential improvements in the site's functional layout, making navigation to key features clearer.

E. Responsiveness: The responsiveness of the site needs attention, especially regarding speed and layout adaptation across devices. Enhancements to reduce lag and ensure smooth adaptations could improve user satisfaction.

F. Navigability: With decent scores, most users can navigate the site intuitively. Continued focus on simplifying navigation can maintain this user-friendly experience.

G. User-Friendliness: Overall, users find the website user-friendly with a pleasant experience. Consistent optimizations in other areas can further enhance overall satisfaction.

5 Listen Phase

Following the Cognitive Walkthrough and survey in Phase 4, initial improvements have been made to key functionality and design, taking into account user feedback. At this stage we reassessed the improvements through an expert review of the site and gained further improvements from the emotional feedback from users to ensure that the functionality and design would continue to be optimized.

5.1 Expert reviews

After implementing adjustments based on the Cognitive Walkthrough, we will conduct expert reviews to ensure these changes effectively enhance the user experience and to identify any remaining design issues.

Expert Review Process:

The expert team will focus on the issues highlighted in the initial Cognitive Walkthrough, specifically testing improvements related to functionality flow, information clarity, and layout logic.

Feedback Collection: Experts will provide detailed feedback on each feature, for example, evaluating whether the adjusted virtual try-on feels more intuitive, collaborative shopping is smoother, and search results are more relevant. Experts will also offer specific suggestions for improving visual aesthetics, information presentation, and interaction design.

Evaluation Area	Specific Feedback Criteria	Details of Expected Feedback
Functionality	Core Feature Usability	Evaluate whether core features (e.g., virtual try-on, image search, collaborative shopping) function as expected and are free from operational issues.
	Interaction Efficiency	Assess how intuitive and efficient each core function is, especially the ease and speed of use for new users.
	Error Handling and Guidance	Check the clarity and helpfulness of error messages and guidance prompts for users when encountering issues.
Layout	Information Hierarchy and Readability	Verify that information is displayed in a logical order, making key features (add to cart, view details) easily accessible and prominent.
	Spacing and Balance	Examine spacing and alignment of elements for visual balance, especially on dense pages (product lists, cart, etc.).
	Adaptability Across Devices	Test layout consistency across devices (desktop, mobile, tablet) to ensure usability is preserved.
3. Content	Product Information Clarity and Detail	Evaluate if product descriptions provide sufficient information (material, size, color) for decision-making.

	Visual and Textual Consistency	Ensure images, icons, and text are cohesive and accurately reflect products, supporting a clear understanding of each item.
	Readability of Text Content	Check font size, style, and contrast to ensure all text content is legible and avoids visual strain.
4. Navigation & User Flow	Ease of Navigation	Assess the clarity and accessibility of navigation menus, ensuring users can find major sections easily.
	User Journey Coherence	Verify that the shopping process flows smoothly without redundant steps or confusing transitions.
	Breadcrumbs and Backtracking	Evaluate the ease of returning to previous pages or modifying selections without losing progress.
5. Visual Design	Color Scheme and Branding	Check if the color scheme aligns with branding goals and enhances the visual hierarchy.
	Iconography and Graphic Elements	Ensure icons are clear, easily understandable, and visually support the corresponding functions without ambiguity.
	Aesthetic Consistency Across Pages	Verify style consistency (buttons, headings) across pages to reinforce a cohesive design.

Figure 32: Table of collecting Expert Reviews Feedback

Iteration Recommendations: Based on expert feedback, we will compile a list of actionable items for further refinement to ensure high standards of usability and simplicity.

5.2 Pay attention to user sentiment

Interviews and questionnaires were designed to further explore users' emotional feedback on the features mentioned in the Cognitive Walkthrough and the expert review. Focus on users' experience with the improved virtual try-on, collaborative chat rooms, image search and natural language heuristic search, and analyze their satisfaction and emotional changes.

We'll analyze users' emotional reactions at each task stage (e.g., satisfaction, confusion, motivation) to identify design or functionality details that evoke positive or negative emotions. For instance, did users find the virtual try-on realistic? Did the collaborative shopping feature make the interaction process feel seamless and enjoyable?

Classify user feedback by emotional response (e.g., satisfaction, frustration, confusion) and create a user sentiment chart to visually map emotional responses across different modules, clarifying priority areas for improvement.

Sentiment Analysis Area	Specific Measurement Methods	Quantifiable Indicators	Expected Insights
Overall User Satisfaction	- Post-interview Likert Scale survey	- Average satisfaction score (1-5 scale)	Understand overall satisfaction levels, pinpointing areas where users feel most positive or negative.
	- Net Promoter Score (NPS)	- NPS score (e.g., “Would you recommend this site?”)	Identify user loyalty and likelihood of recommending the platform.
Emotional Response	- In-situ emotional reaction recording	- Percentage of positive vs. negative reactions	Measure initial emotional reactions to features, especially new or unique ones like virtual try-on.
	- Post-task feedback (e.g., “How did using [feature] feel?”)	- Score indicating positive or negative sentiment	Capture emotional feedback for specific features, such as ease of use, frustration, or enjoyment.
Frustration Points	- Open-ended feedback on pain points	- Frequency of negative comments by feature	Identify features or actions that cause user frustration, and assess frequency of issues.
	- Cognitive Walkthrough notes	- Number of errors/missteps per user task	Quantify areas where users commonly struggle, revealing high-frustration areas.
Engagement and Ease of Use	- Session recording analysis (e.g., time spent per feature)	- Average time on key features	Gauge engagement with each feature, identifying features with high or low engagement.
	- Drop-off rates from feature to feature	- Completion rate of tasks (e.g., completing purchase)	Understand the ease of progression through tasks and user journey flow; high drop-off points indicate friction.
Feature Usefulness	- Likert scale ratings (e.g., usefulness of virtual try-on)	- Average usefulness rating	Quantify perceived usefulness of individual features, distinguishing between high and low-value functions.
	- Open feedback on functionality effectiveness	- Specific mentions of beneficial or redundant features	Gather user perceptions on which features add real value vs. those needing improvement or removal.
Emotional Engagement	- Positive/negative keyword analysis from interviews	- Frequency of positive/negative emotional terms	Identify emotional terms associated with features to see if users feel connected or distanced by them.
User Journey Satisfaction	- After-task Likert scale on ease of completion	- Task ease rating (1-5)	Quantify satisfaction with the navigation and flow, assessing if each step in the user journey feels smooth.
	- Sentiment scoring for each step (e.g., checkout process)	- Sentiment trend across steps	Detect which parts of the user journey enhance or hinder the overall experience.

Figure 33: table of collecting user emotional reactions

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7 Appendix

7.1 Mind Map

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7.2 Questionnaire Quantitate Question

<https://forms.gle/tev8ERvBKtkxNWzT6>

7.3 Questionnaire Qualitative Question

<https://docs.google.com/forms/d/1dESoJSEE0NzSYChNKgcFNNVRGW0Xd8qjDy3Ek2e3cVY/edit>

7.4 Survey Quantitative-Qualitative Table: Questionnaire Survey Statistics and Analysis

https://tudublin-my.sharepoint.com/:x/r/personal/d23125391_mytudublin_ie/_layouts/15/Doc.aspx?sourcedoc=%7B36F5F1D7-3641-4942-817A-97C42EB2EBF9%7D&file=Survey%20Quantitative-Qualitative%20Table.xlsx&action=default&mobileredirect=true